

what is a
Database?



what's the
problem
they solve?



Databases are used to
store any kind of data



why Databases? and not normal file systems?



Google Sheets



size of the data



size of the data
ease of updating



size of the data
ease of updating
duplication



size of the data
ease of updating
duplication
importance



size of the data
ease of updating
duplication
importance
secure / confidential

CONFIDENTIAL



a database is
structured system
governed by **rules**
that you can set based
on your **personal**
customized needs





DBMS



DBMS

a software used to manage one or several databases each having different set of rules and data stored in them



Relational DBMS

Hierarchical DBMS

Network DBMS

Object-Oriented
DBMS

NoSQL DBMS



column

Game	Name	Genre	ReleaseDate
	World of Warcraft	MMORPG	23/11/2004
	Assassin's Creed: Valhalla	Action Role Playing	12/11/2020

row

...

first step is to define the table and its columns, and only after doing that, we can go ahead and insert our rows



primary key

Game	Name	Genre	ReleaseDate
	World of Warcraft	MMORPG	23/11/2004
	Assassin's Creed: Valhalla	Action Role Playing	12/11/2020

...

the primary key identifies each record as unique inside our table



primary key

Game	Name	Genre	ReleaseDate
	World of Warcraft	MMORPG	23/11/2004
	Assassin's Creed: Valhalla	Action Role Playing	12/11/2020
	Diablo	Action Role Playing	27/6/2017
	Diablo	Mobile Game	N/A

“Name” is no longer a valid primary key

...

the primary key identifies each record as unique inside our table



primary key

Game	GameId	Name	Genre	ReleaseDate
	1	World of Warcraft	MMORPG	23/11/2004
	2	Assassin's Creed: Valhalla	Action Role Playing	12/11/2020
	3	Diablo	Action Role Playing	27/6/2017
	4	Diablo	Mobile Game	N/A

...

this column is still a primary key, but you may also refer to it as synthetic key or surrogate key



Game	GameId	Name	Genre	ReleaseDate
	1	World of Warcraft	MMORPG	23/11/2004
	2	Assassin's Creed: Valhalla	Action Role Playing	12/11/2020
	3	Diablo	Action Role Playing	27/6/2017
	4	Diablo	Mobile Game	N/A

we only have one game with id = 1 for example



let's say that in our database we have three tables:

the 'Game' table that we already saw,
which stores the details of given games

the 'Company' table, which stores information
related to several gaming companies

and the 'Streamer' table, which stores the
information related to several game streamers



Game	GameId	Name	Genre	ReleaseDate
	1	World of Warcraft	MMORPG	23/11/2004
	2	Assassin's Creed: Valhalla	Action Role Playing	12/11/2020
	3	Marvel Future Fight	Mobile Game	30/4/2015
	4	Seven Deadly Sins	Mobile Game	3/3/2020

Streamer	StreamerId	Name	...
	1	Asmongold	...
	2	JordanDoes	...
	3	Cynicalalex	...

Company	CompanyId	Name	Headquarters	...
	1	Blizzard Entertainment	California, U.S.	...
	2	Ubisoft	Montreuil, France	...
	3	Netmarble	Seoul, South Korea	...



how do we represent relationships between tables?

every game is developed by one and only one company,
but how do we represent this in our tables?

instead of duplicating all company info next to each game,
we add the corresponding 'companyId' to our Game table

this column is a foreign key, defining a one-to-many
relationship: one company develops many games, but
each game belongs to one company



Game	GameId	Name	Genre	ReleaseDate
	1	World of Warcraft	MMORPG	23/11/2004
	2	Assassin's Creed: Valhalla	Action Role Playing	12/11/2020
	3	Marvel Future Fight	Mobile Game	30/4/2015
	4	Seven Deadly Sins	Mobile Game	3/3/2020

Company	CompanyId	Name	Headquarters	...
	1	Blizzard Entertainment	California, U.S.	...
	2	Ubisoft	Montreuil, France	...
	3	Netmarble	Seoul, South Korea	...



Game	GameId	Name	Genre	ReleaseDate	CompanyName	Headquarters	...
	1	World of Warcraft	MMORPG	23/11/2004	Blizzard Entertainment	California, U.S.	...
	2	Assassin's Creed: Valhalla	Action Role Playing	12/11/2020	Ubisoft	Montreuil, France	...
	3	Marvel Future Fight	Mobile Game	30/4/2015	Netmarble	Seoul, South Korea	...
	4	Seven Deadly Sins	Mobile Game	3/3/2020	Netmarble	Seoul, South Korea	...

Company	CompanyId	Name	Headquarters	...
	1	Blizzard Entertainment	California, U.S.	...
	2	Ubisoft	Montreuil, France	...
	3	Netmarble	Seoul, South Korea	...

this duplicates company info
next to every game
unnecessary and error-prone



Game	GameId	Name	Genre	ReleaseDate	CompanyId
	1	World of Warcraft	MMORPG	23/11/2004	1
	2	Assassin's Creed: Valhalla	Action Role Playing	12/11/2020	2
	3	Marvel Future Fight	Mobile Game	30/4/2015	3
	4	Seven Deadly Sins	Mobile Game	3/3/2020	3

Company	CompanyId	Name	Headquarters	...
	1	Blizzard Entertainment	California, U.S.	...
	2	Ubisoft	Montreuil, France	...
	3	Netmarble	Seoul, South Korea	...



Game	GameId	Name	Genre	ReleaseDate	CompanyId
	1	World of Warcraft	MMORPG	23/11/2004	1
	2	Assassin's Creed: Valhalla	Action Role Playing	12/11/2020	2
	3	Marvel Future Fight	Mobile Game	30/4/2015	3
	4	Seven Deadly Sins	Mobile Game	3/3/2020	3

Company	CompanyId	Name	Headquarters	...
	1	Blizzard Entertainment	California, U.S.	...
	2	Ubisoft	Montreuil, France	...
	3	Netmarble	Seoul, South Korea	...

this column is not unique in the "Game" table



Game	GameId	Name	Genre	ReleaseDate	CompanyId
	1	World of Warcraft	MMORPG	23/11/2004	1
	2	Assassin's Creed: Valhalla	Action Role Playing	12/11/2020	2
	3	Marvel Future Fight	Mobile Game	30/4/2015	3
	4	Seven Deadly Sins	Mobile Game	3/3/2020	3

foreign
key

Company	CompanyId	Name	Headquarters	...
	1	Blizzard Entertainment	California, U.S.	...
	2	Ubisoft	Montreuil, France	...
	3	Netmarble	Seoul, South Korea	...

One-to-Many
relationship



what about many-to-many relationships?

a game can be streamed by one or several streamers,
and a streamer can stream one or several games

a single foreign key column won't work here, as we
need to represent multiple values on both sides

instead, we create a junction (or linking) table whose
only purpose is to join the two tables together



Game	GameId	Name	Genre	ReleaseDate	...
	1	World of Warcraft	MMORPG	23/11/2004	...
	2	Assassin's Creed: Valhalla	Action Role Playing	12/11/2020	...
	3	Marvel Future Fight	Mobile Game	30/4/2015	...
	4	Seven Deadly Sins	Mobile Game	3/3/2020	...

Streamer	StreamerId	Name	...
	1	Asmongold	...
	2	JordanDoes	...
	3	Cynicalalex	...



Game	GameId	Name	Genre	ReleaseDate	...
	1	World of Warcraft	MMORPG	23/11/2004	...
	2	Assassin's Creed: Valhalla	Action Role Playing	12/11/2020	...
	3	Marvel Future Fight	Mobile Game	30/4/2015	...
	4	Seven Deadly Sins	Mobile Game	3/3/2020	...

Many-to-Many
relationship

Streamer	StreamerId	Name	...
	1	Asmongold	...
	2	JordanDoes	...
	3	Cynicalalex	...



Game	GameId	Name	Genre	ReleaseDate	...
	1	World of Warcraft	MMORPG	23/11/2004	...
	2	Assassin's Creed: Valhalla	Action Role Playing	12/11/2020	...
	3	Marvel Future Fight	Mobile Game	30/4/2015	...
	4	Seven Deadly Sins	Mobile Game	3/3/2020	...

Many-to-Many
relationship

Streamer	StreamerId	Name	GameId
	1	Asmongold	1
	2	JordanDoes	2
	3	Cynicallex	?

streams
two games



Game	GameId	Name	Genre	ReleaseDate	...
	1	World of Warcraft	MMORPG	23/11/2004	...
	2	Assassin's Creed: Valhalla	Action Role Playing	12/11/2020	...
	3	Marvel Future Fight	Mobile Game	30/4/2015	...
	4	Seven Deadly Sins	Mobile Game	3/3/2020	...

Many-to-Many relationship

both approaches are highly discouraged – comma-separated values and extra columns break database design rules

Streamer	StreamerId	Name	GameId
	1	Asmongold	1
	2	JordanDoes	2
	3	Cynicallex	3, 4

OR

Streamer	StreamerId	Name	GameId 1	GameId 2
	1	Asmongold	1	null
	2	JordanDoes	2	null
	3	Cynicallex	3	4



Game	GameId	Name	Genre	ReleaseDate	...
	1	World of Warcraft	MMORPG	23/11/2004	...
	2	Assassin's Creed: Valhalla	Action Role Playing	12/11/2020	...
	3	Marvel Future Fight	Mobile Game	30/4/2015	...
	4	Seven Deadly Sins	Mobile Game	3/3/2020	...

**Many-to-Many
relationship**

Streamer	StreamerId	Name	...
	1	Asmongold	...
	2	JordanDoes	...
	3	Cynicalalex	...

**junction or
linking table**

GameStreamer	GameId	StreamerId
	1	1
	2	2
	3	3
	4	3



Game	GameId	Name	Genre	ReleaseDate	...
	1	World of Warcraft	MMORPG	23/11/2004	...
	2	Assassin's Creed: Valhalla	Action Role Playing	12/11/2020	...
	3	Marvel Future Fight	Mobile Game	30/4/2015	...
	4	Seven Deadly Sins	Mobile Game	3/3/2020	...

Streamer	StreamerId	Name	...
	1	Asmongold	...
	2	JordanDoes	...
	3	Cynicalalex	...

GameStreamer	GameId	StreamerId
	1	1
	2	2
	3	3
	4	3



the third kind: one-to-one relationships

if a row in one table points to one and only one row in another table, and vice versa

this is kind of pointless, because we might as well combine these two tables into one

still, one-to-one relationships exist and are sometimes used to split large tables for performance or organizational reasons



Game	GameId	Name	Genre	ReleaseDate	GameDetails
	1	World of Warcraft	MMORPG	23/11/2004	1
	2	Assassin's Creed: Valhalla	Action Role Playing	12/11/2020	2
	3	Marvel Future Fight	Mobile Game	30/4/2015	3
	4	Seven Deadly Sins	Mobile Game	3/3/2020	4

GameDetails	DetailsId	Size (GBs)	Players	...
	1	101	100M	...
	2	60	10M	...
	3	2.3	100M	...
	4	1	1M	...

**One-to-One
relationship**



what if we need to change data in our tables?

updating values is common, but what happens when multiple people try to change the same data at the same time?

we need a mechanism that ensures either all operations succeed together, or none of them execute at all

this is exactly what transactions allow us to do



Transactions

BankAccount	AccountId	Balance	Type	...
	1	3 000	Payroll	...

You:

- Read value from DB
- Subtract 1000
- Update value to DB

Wife:

- Read value from DB
- Subtract 1000
- Update value to DB



Transactions

BankAccount	AccountId	Balance	Type	...
	1	4 000	Payroll	...



You:

- Read value from DB (5000)
- Subtract 1000 (4000)
- Update value to DB (4000)

Wife:

- Read value from DB (5000)
- Subtract 1000 (4000)
- Update value to DB (4000)



Transactions

BankAccount	AccountId	Balance	Type	...
	1	3 000	Payroll	...



Start Transaction:

- Read value from DB (5000)
- Subtract 1000 (4000)
- Update value to DB (4000)

End

Start Transaction:

- Read value from DB (4000)
- Subtract 1000 (3000)
- Update value to DB (3000)

End



Transactions

Attomic happens completely or none
of it gets executed at all

Consistent rules we set in our database cannot
be defied inside a transaction

Isolated data is locked to avoid multiple
updates at the same time

Durable the transaction guarantees
a successful message



Structured Query Language

declarative
query language

Game	GameId	Name	Players
	1	World of Warcraft	100M
	2	Assassin's Creed	10M
	3	Marvel Future Fight	100M
	4	Seven Deadly Sins	1M

Imperative Language:

```
List<Game> highPlayerBase = new ArrayList<>();  
for (Game game : gameTable.entries()) {  
    if (game.getPlayers() > 10_000_000) {  
        highPlayerBase.add(game);  
    }  
}  
return highPlayerBase;
```

Declarative Language:

fetch for me the games having a number of
players of more than ten million

```
SELECT * FROM Game WHERE Players > 10000000
```



Structured Query Language



Create

Read

Update

Delete



Recap

a database is not just a place to store data – it is a structured system governed by rules you set based on your needs

it keeps your data secure, accurate, and accessible – while scaling to millions of records and hundreds of users

next up: everything you need to know about SQL

